

SMART INDIA HACKATHON 2026



- Problem Statement ID – 26037
- Problem Statement Title- ADAPTIVE PATH PLANNING AND COLLISION AVOIDANCE FOR AUTONOMOUS VEHICLES ON UNSTRUCTURED INDIAN ROADS
- Theme- SAFE TRANSPORTATION AND SMART MOBILITY
- PS Category- Software
- Team ID- 139838
- Team Name(Registered on portal)- ByteRoute AV

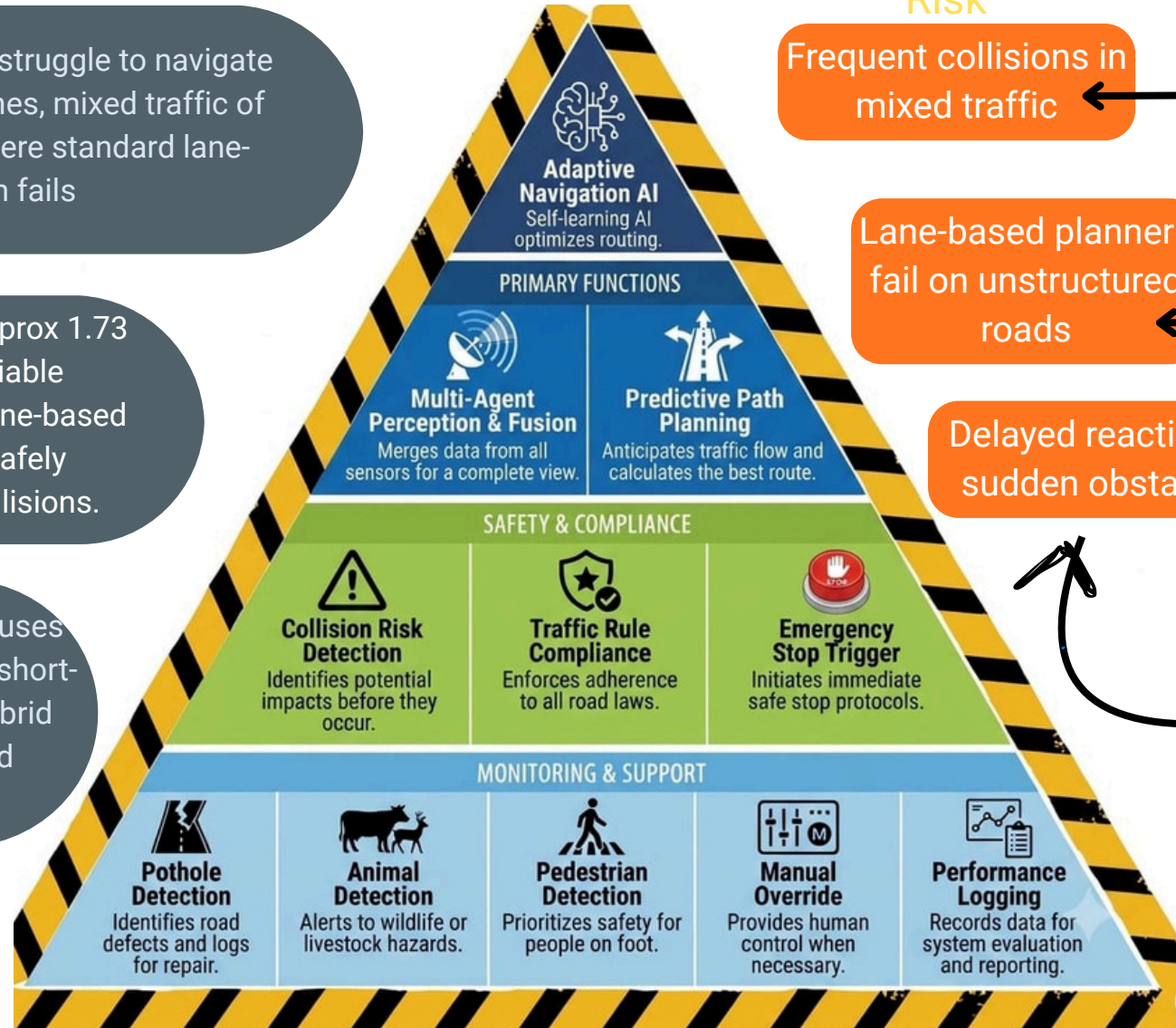


Adaptive Path: AI-Driven Perception and Planning for safe Autonomous Navigation on Unstructured Roads

Real World Issues: Autonomous vehicles struggle to navigate on unstructured Indian roads (no clear lanes, mixed traffic of cars, pedestrians, animals, pushcarts) where standard lane-based self driving algorithm fails

Why important: Indian roads claim over approx 1.73 lakhs lives per year, so without any reliable perception and planning for chaotic non-lane-based traffic, autonomous vehicles can't be safely deployed in Indian conditions, risking collisions.

Solution: A simulated SIMULINK pipeline that fuses sensor data to detect road users, predicts their short-term motion, and plans collision-free paths (Hybrid A*/RRT*) with real-time replanning for mixed, unstructured traffic.



Risk

Frequent collisions in mixed traffic

Lane-based planners fail on unstructured roads

Delayed reaction to sudden obstacles

Solution

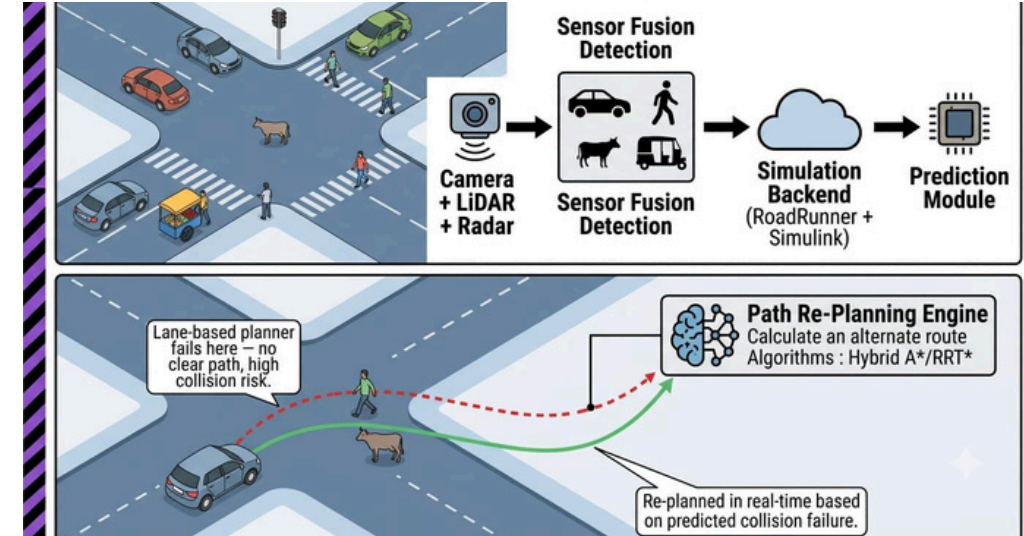
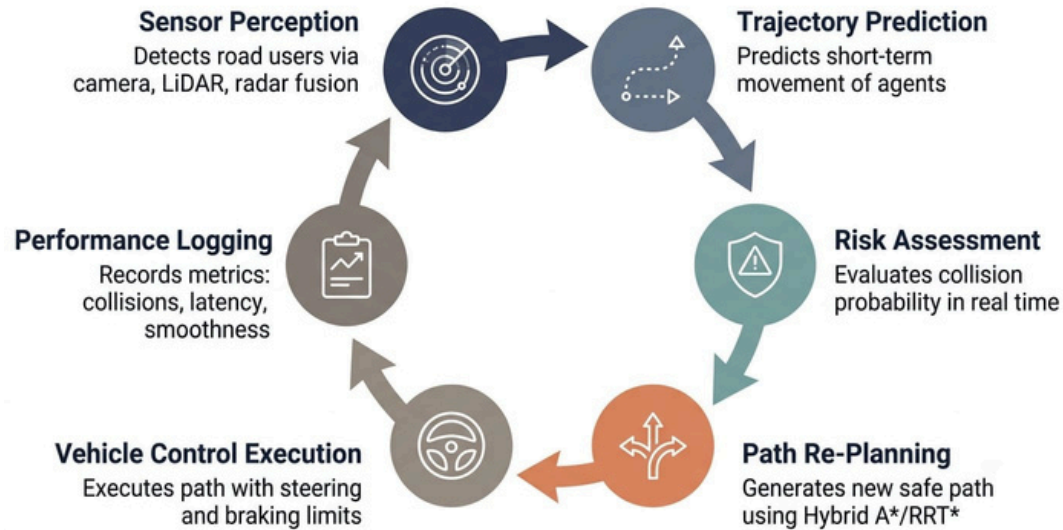
Real time perception + predictive path planning

Hybrid A*/RRT non-lane navigation

Reactive real time re-planning



METHODOLOGY & PROCESS OF IMPLEMENTATION



TECHNOLOGIES USED

FrontEnd
React/CSS

BackEnd
Supabase model

Deployment
Simulation environment which uses MATLAB/SIMULINK

Software
Simulated camera /RADAR/LIDAR

[Click for Video](#)

They're for an optional results dashboard, not the core system.

Feasibility:

High — MATLAB's Automated Driving Toolbox, Sensor Fusion & Tracking Toolbox, and Vehicle Dynamics Blockset cover perception, Hybrid A* planning, and control out of the box. RoadRunner builds Indian edge cases via waypoints, and a standard laptop hits <50ms replanning for 2D kinematic scenarios.

Risks → Fixes:

3D rendering freezes live demos → use 2D boxes for scoring, save 3D for pre-recorded video
Erratic agents break Kalman prediction → class-based safety bubbles (1.5x buffer for animals/autos)
Dense obstacles cause deadlocks → Hybrid A* + MPC, with nudge/stop fallback after 2s stuck
RoadRunner–Simulink integration breaks under time pressure → modify MathWorks' existing Parking/Trajectory examples instead of building from scratch

Viability:

High — this isn't a hypothetical problem. Bengaluru-based Minus Zero is developing an end-to-end ADAS and autonomous driving stack tailored to India's unstructured traffic conditions, and Kochi-based RoshAi is building a full-stack autonomy approach with a sensor-fusion pipeline for on-road use. Separately, ARAI has been testing LiDAR-equipped vehicles to evaluate self-driving feasibility on Indian roads. Real companies and regulators are already investing in exactly this problem.

Why it holds up beyond the demo:

Low-risk validation path — proving the pipeline in RoadRunner/Simulink before real-world testing mirrors how the industry itself de-risks autonomy (simulate first, road-test later)

Generalizes past the two required scenes — the same perception → prediction → planning loop extends to any unstructured road type (rural, monsoon, hill roads) without redesigning the core architecture

Real precedent for impact — comparable AI collision-avoidance systems already show 43–64% real-world crash reduction, giving this approach a credible ceiling once matured past simulation

Business Models:

Government Cost Savings
Cuts costly real-world AV trial expenses by validating logic in simulation first
Low-cost input for road-safety/infrastructure planning (flagging high-risk unstructured zones)
New Revenue Generation
Licensable benchmark/certification tool for AV companies building India-specific autonomy stacks
Sellable RoadRunner scenario dataset (village road, urban intersection) for other developers to test against

IMPACTS

Reduced Collisions:

Designed to substantially cut collisions in unstructured mixed-traffic zones through predictive path planning. Comparable AI collision-avoidance systems show 43-64% crash reduction in published studies - our target, pending validation.

Faster, Safer Emergency Responset:

Removes human reaction-delay bottleneck via real-time replanning (<50ms design target) instead of relying on driver response time.

Lower Accident-Related Losses:

Cuts costs from vehicle damage, medical expenses, and traffic-related fatalities.

Real-Time Hazard Alerts:

Detects potholes, animals, and jaywalking pedestrians before they cause incidents.

BENEFITS

1. Safety

Collision-free navigation validated across 5+ real-world-style scenarios (village roads, urban intersections).

2. Scalability

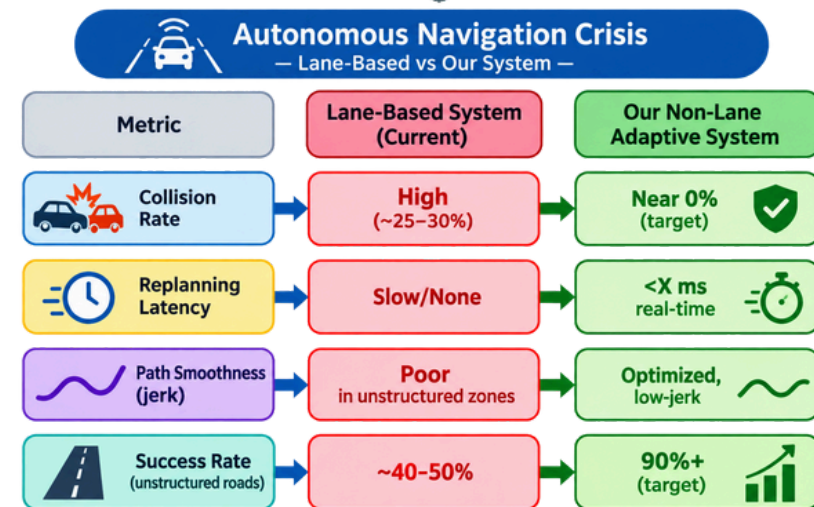
Works on any unstructured road — no dependency on lane markings, unlike conventional AV planners.

3. Environmental

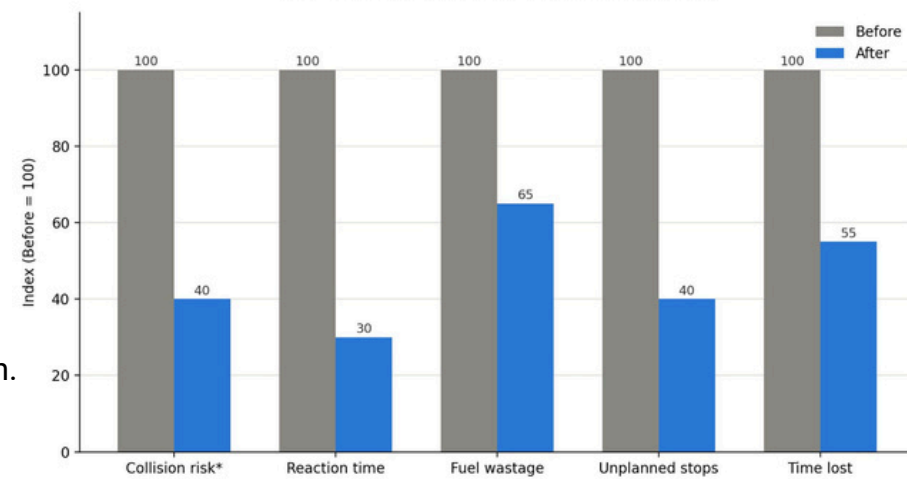
Smoother paths (lower jerk/curvature) reduce fuel wastage and emissions from erratic braking/acceleration.

4. Technology

AI-driven perception, prediction, and reactive re-planning — adaptable to Indian road conditions at scale.



Collision avoidance: before vs after (illustrative)



*Collision Risk is anchored to cited data (pedestrian fatality share, AEB collision-reduction studies). The other four bars are directional estimates for illustration, not measured results.

NITI Aayog:
Smart Cities &
Infrastructure
<https://www.niti.gov.in/>

MoRTH:
Road Accidents
in India Report
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IEEE Xplore:
Retrieve active
critical and
historical
information
<https://ieeexplore.ieee.org/document/11378941>

MathWorks:
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Driving Toolbox
for navigation
<https://in.mathworks.com/>

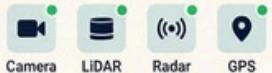


ByteRoute AV: Adaptive Path Planning for Unstructured Indian Roads

Multi-sensor fusion, adaptive
multisensor fusion and real-time
collision avoidance.

View Simulation

Learn More



Approach Comparison: Handling Unstructured Road Challenges

■ Camera-Only System ■ LiDAR+Camera System ■ Standard OEM ADAS ■ Generic Adaptive System

